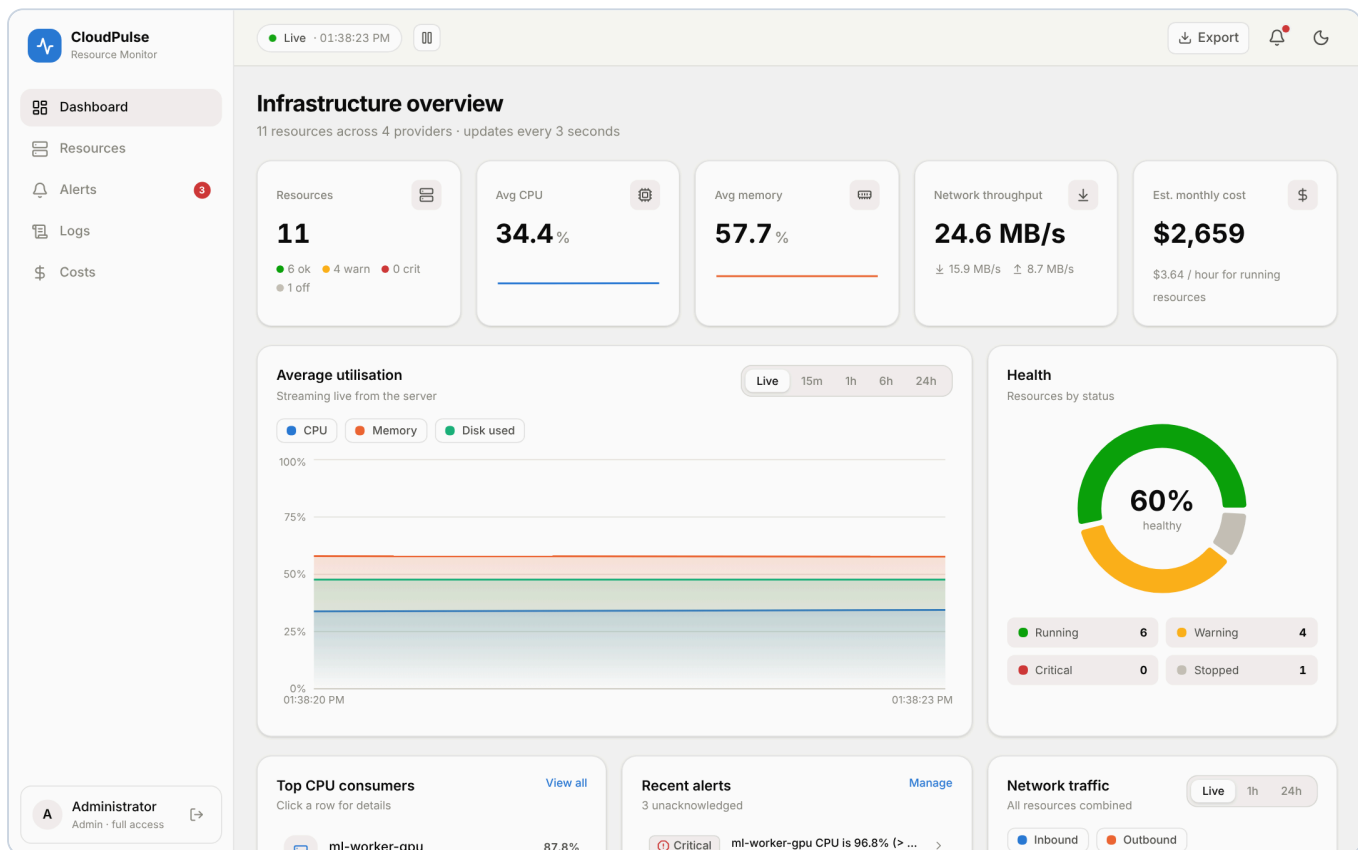


CloudPulse User Manual

How to install, start and use the Cloud Resource Monitoring Dashboard



Software: CloudPulse – Cloud Resource Monitoring Dashboard

Developed by: Cecil Baraik • Roll 2025-BSCIT/004A

Works with: Java 21 • MySQL 8 • Node.js • Chrome or Edge

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About CloudPulse

CloudPulse is a web application that shows the health of cloud servers on one live screen. It updates every 3 seconds, raises alerts when something goes wrong, sends email for serious problems, shows what the servers cost, shows live logs and creates PDF and CSV reports.

It watches **11 resources**: 10 demo cloud servers on AWS, Azure and Google Cloud (simulated) and **your own computer** (real data).

Who uses it

Role	Demo login	What they can do
Admin	admin / admin123	Everything: watch, start/stop servers, edit alert rules, change budget and email settings.
Viewer	viewer / viewer123	Watch everything, acknowledge alerts, export reports. Cannot change anything.

Words used in this manual

Resource	One monitored item: a server, database, container, storage, function, or your PC.
CPU / Memory / Disk used	How busy the processor is, how full the RAM is, and how full the storage space is, in %.
Network in / out	Data received / sent per second.
Alert	A warning created when a value crosses a limit set in a rule .
Acknowledge	Mark an alert as "seen, I am handling it".
Status colours	● Running (all below 75%) · ● Warning (75–89%) · ● Critical (90% or more) · ● Stopped

Installation and first-time setup

2.1 What you need

Software	Version	Why
Java JDK	21	Runs the backend server
MySQL Server	8.x	Stores all data
Node.js (with npm)	20 or newer	Runs the website (frontend)
IntelliJ IDEA	any recent	To open and run the backend
Google Chrome / Edge	any recent	To use CloudPulse
Gmail account	optional	Only for email alerts

2.2 Project folders

```
cloud-monitor\  
  backend\    ← Java Spring Boot server (open this in IntelliJ)  
  frontend\   ← React website
```

2.3 Set up the database (one time)

1. Install MySQL 8 and remember the **root** password.
2. Open `backend\src\main\resources\application.properties` and check the password line:

```
spring.datasource.password=${DB_PASSWORD:root123}
```

If your MySQL root password is not `root123`, either change it here or set an environment variable `DB_PASSWORD`.

3. You do **not** need to create the database or tables. CloudPulse creates the database `cloudmonitor` and all 6 tables automatically on first start, and fills them with demo data.

2.4 Install the website packages (one time)

Open PowerShell in the `frontend` folder and run:

```
npm install
```

This downloads the libraries into `node_modules`. You only do this once (or after a library is added).

2.5 Email alerts (optional, one time)

See **Section 10**. You can use everything else without email.

Starting and stopping CloudPulse

Always start in this order: 1. MySQL → 2. Backend → 3. Frontend → 4. Browser

3.1 Start

1. **MySQL:** it normally starts with Windows. To check: press Win + R, type `services.msc`, find **MySQL80** and make sure it says *Running*.
2. **Backend:** open the `backend` folder in IntelliJ and click the green ▶ Run button next to `CloudMonitorApplication`. Wait until the Run window shows:

```
Started CloudMonitorApplication in ... seconds  
Email alerts: ready ... (only if email is set up)
```

3. **Frontend:** open PowerShell in the `frontend` folder and run:

```
npm run dev
```

Wait until it shows `Local: http://localhost:5173/`.

4. **Browser:** open `http://localhost:5173`.

3.2 Stop

1. In PowerShell press **Ctrl + C** to stop the frontend.
2. In IntelliJ click the red ■ Stop button.
3. MySQL can keep running. All data is saved, so next time everything is still there.

If the browser shows "Offline" or empty pages, the backend is not running. Start it and the page reconnects by itself within a few seconds.

Logging in and out

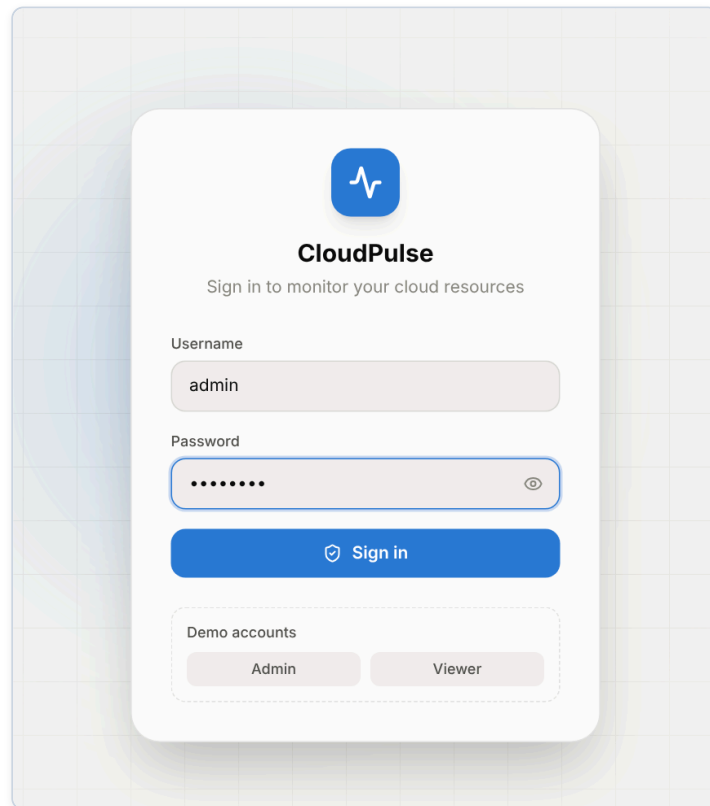


Figure 1 – Login screen

1. Type your **Username** and **Password**, then click **Sign in** (or press Enter).
 2. To fill the demo accounts quickly, click **Admin** or **Viewer** under *Demo accounts*.
 3. Click the **eye** icon to show or hide the password.
- Wrong details show "Invalid username or password".
 - Your login stays valid for **8 hours**. After that you are sent back to the login screen automatically.
 - **Log out**: click the arrow icon next to your name at the bottom of the left menu.

The screen layout

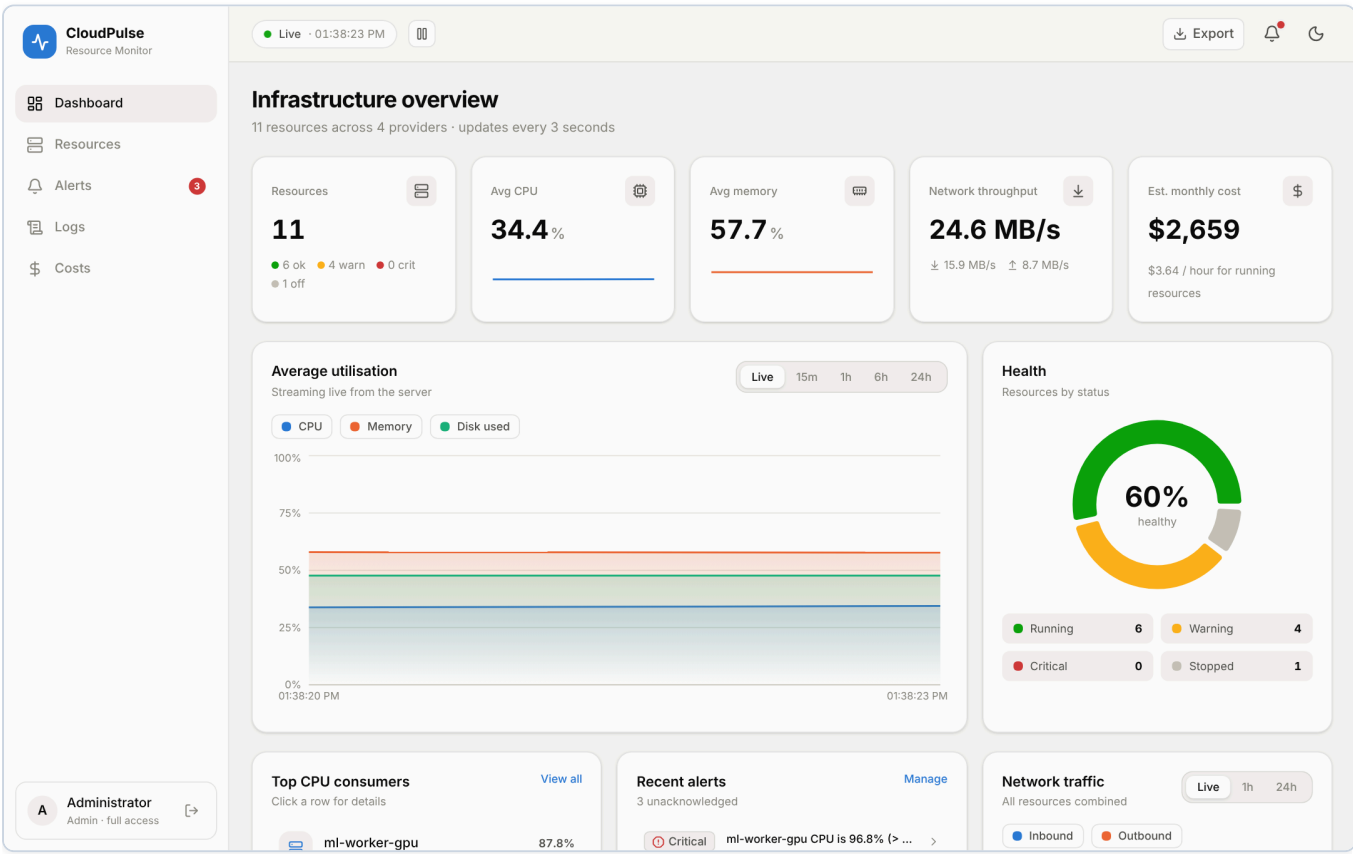


Figure 2 – Main screen with left menu and top bar

Part	What it does
Left menu	Dashboard, Resources, Alerts (red number = active alerts), Logs, Costs.
Live • time	Green dot = connected and receiving data; the time of the last update.
Pause ■	Freezes the live charts so you can read them. Click again to resume.
Export	Download the PDF report or CSV files (Section 13).
Bell 🛎	Opens the Alerts page. A red dot means there are unacknowledged alerts.
Moon / Sun	Switch between dark and light theme.
Your name (bottom left)	Shows who is logged in and the role; the arrow logs out.

Dashboard

The first page after login. It gives the overall picture in one look.

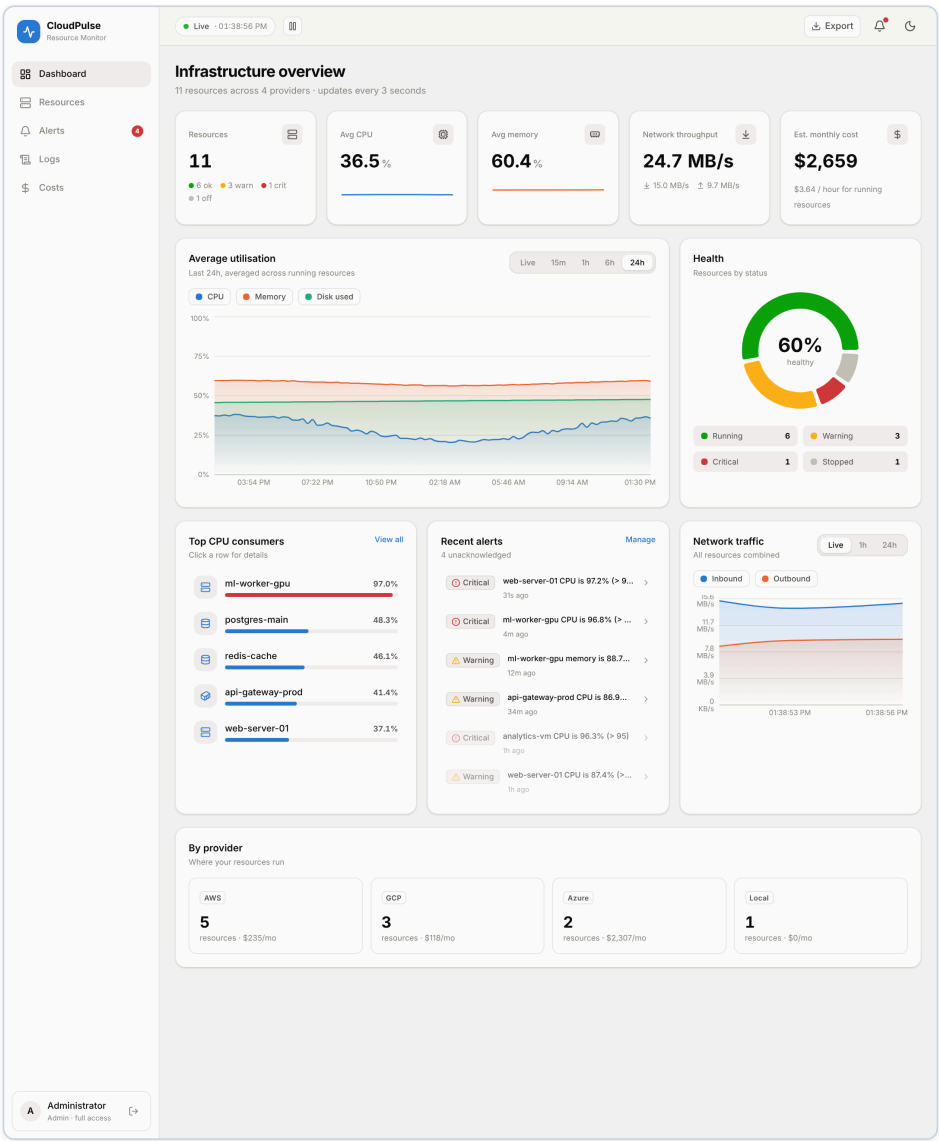


Figure 3 – Dashboard (24h range selected)

Area	How to read it
Top cards	Number of resources (ok / warn / crit / off), average CPU, average memory, total network speed, estimated monthly cost.
Average utilisation	CPU, memory and disk of all running resources. Choose Live (last few minutes, moving), 15m , 1h , 6h , 24h (history). Click a legend item (CPU / Memory / Disk used) to hide or show that line.
Health	Ring showing how many resources are Running, Warning, Critical, Stopped. The % in the middle is the share that is healthy.
Top CPU consumers	Busiest resources first. Click a row to open its details. <i>View all</i> opens Resources.
Recent alerts	Latest alerts. <i>Manage</i> opens the Alerts page.

Network traffic Inbound (blue) and outbound (orange) data for all resources.

By provider How many resources are on AWS, GCP, Azure and Local, and what they cost.

Tip: Hover the mouse over any chart to see the exact values at that time.

Resources

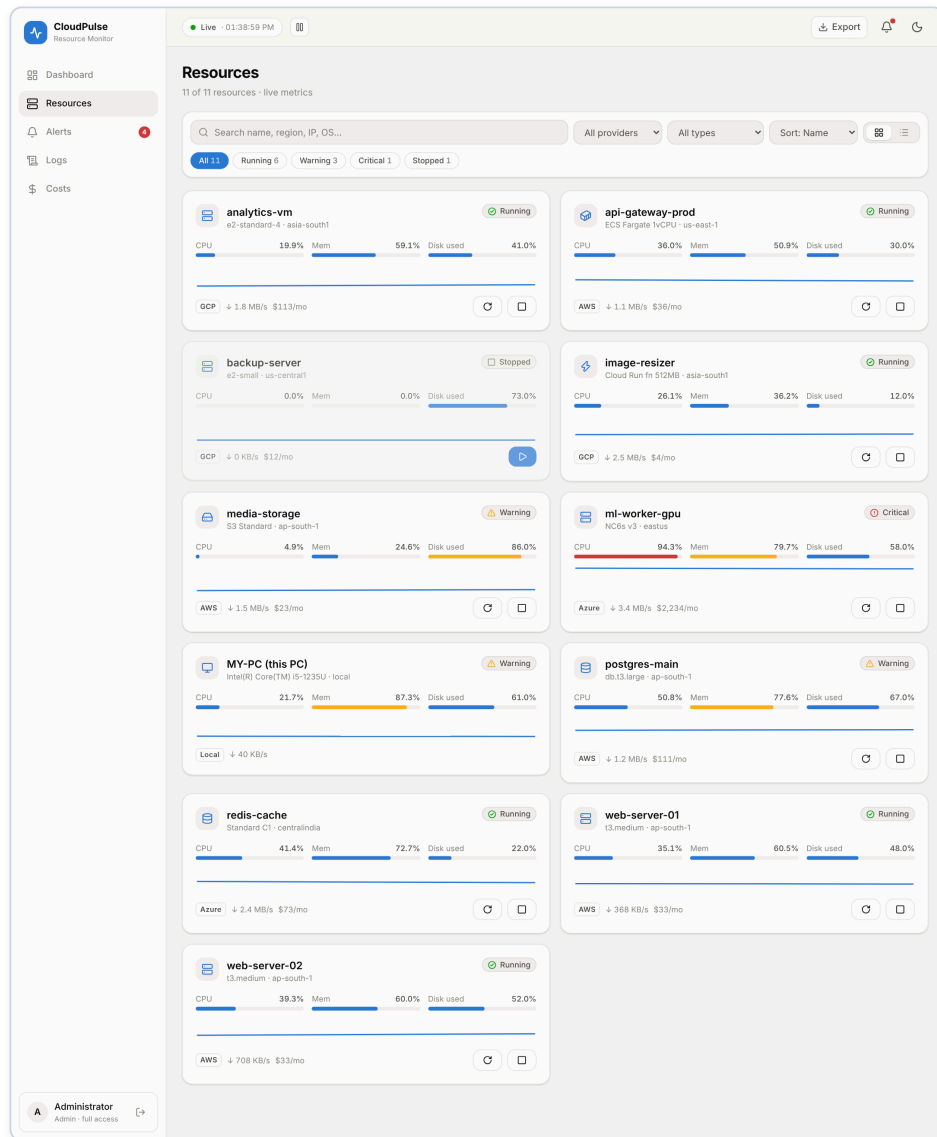


Figure 4 – Resources page (card view)

7.1 Find a resource

- **Search box:** type part of a name, region, IP address or OS (e.g. **web**, **ap-south-1**, **10.0.1** or **Ubuntu**).
- **Filters:** choose a provider, a type, and a status (All, Running, Warning, Critical, Stopped).
- **Sort:** by name, CPU, memory, disk or cost.
- **View buttons** (top right): **cards** or **table**. In table view you can click column headings to sort. Your choice is remembered.

7.2 What each card shows

Name and provider, status badge, current CPU, memory and disk bars, a small trend line (sparkline), network speed and monthly cost. Click a card to open the details page.

Resource details, start and stop

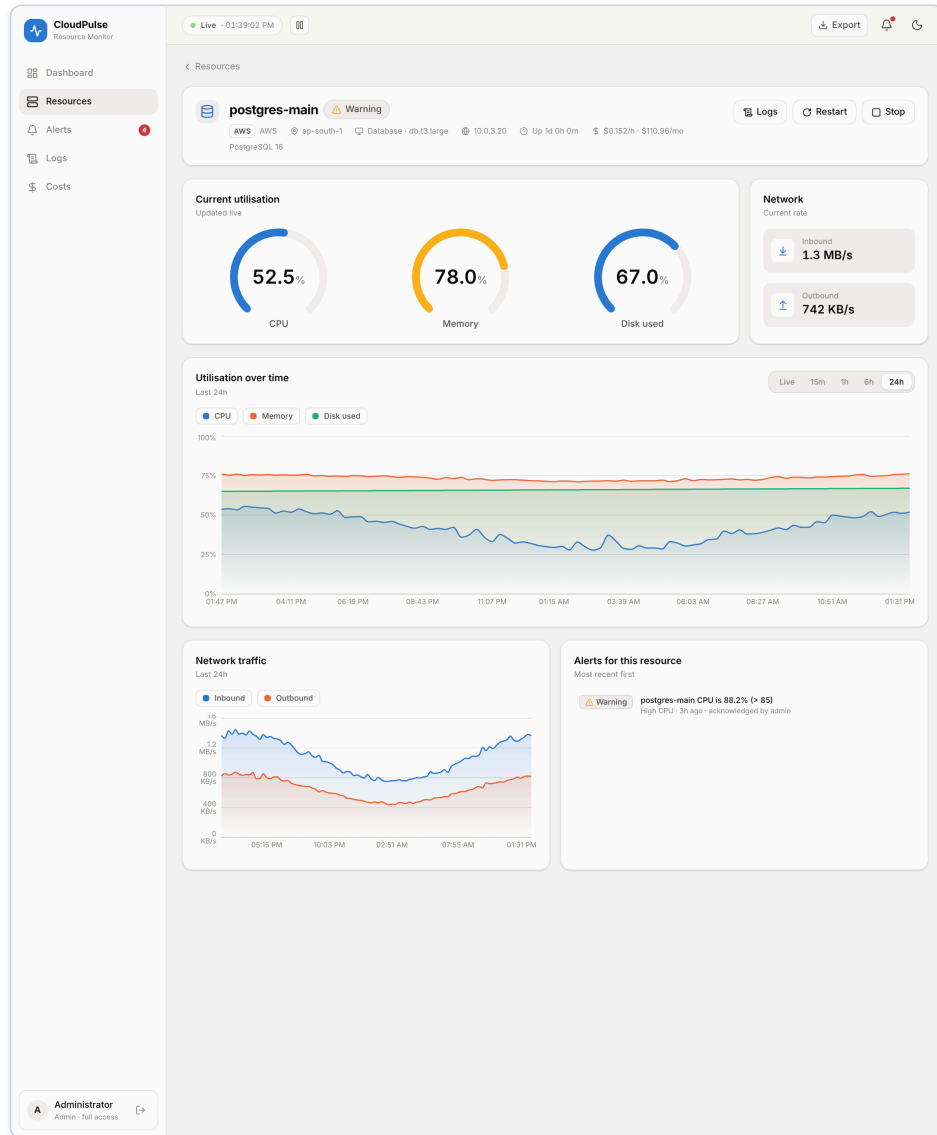


Figure 5 – Details of one resource (postgres-main, 24h)

8.1 What you see

- **Header:** name, status, provider, region, type and size, IP address, uptime ("Up 3h 0m"), price per hour and per month, operating system.
- **Current utilisation:** three gauges for CPU, memory and disk used. Blue = normal, amber = warning, red = critical.
- **Network:** current inbound and outbound speed.
- **Utilisation over time:** Live, 15m, 1h, 6h or 24h.
- **Alerts for this resource:** its recent alerts.
- **Logs** button: opens the Logs page showing only this resource.

8.2 Start, stop and restart (admin only)

- 1. Open the resource and click **Stop**, **Start** or **Restart** (top right).
- 2. A confirmation box appears. Click the coloured button to confirm, or **Cancel**.

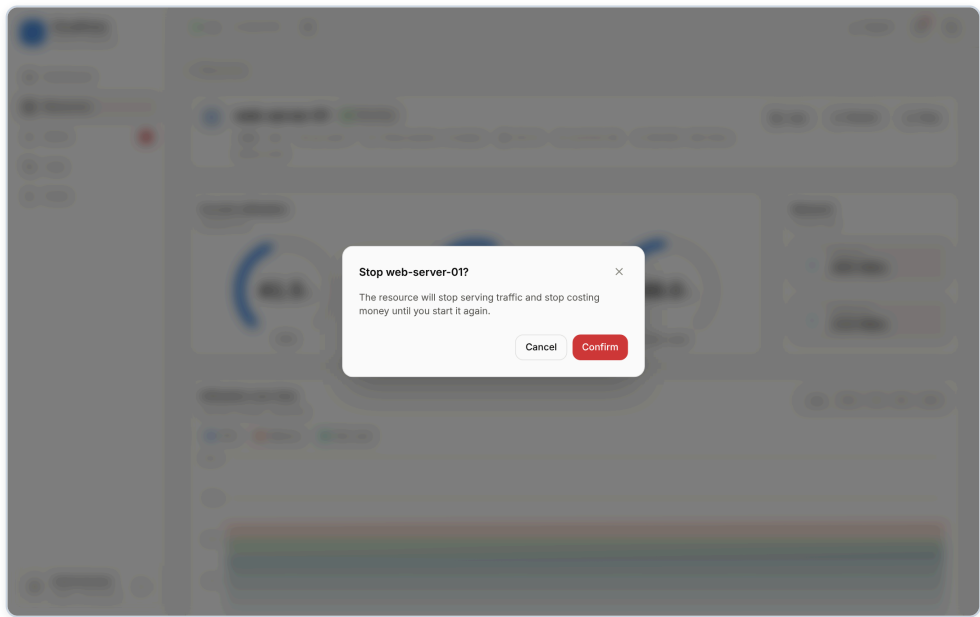


Figure 6 – Confirm before stopping

Action	Result
Stop	Status becomes Stopped (grey). It uses 0% and costs nothing. No alerts for it.
Start	Becomes Running again; uptime starts from now.
Restart	Same as start; clears any load spike.

Every action is recorded as a **System event** alert with your name (e.g. "backup-server was stopped by admin").

8.3 Your own PC

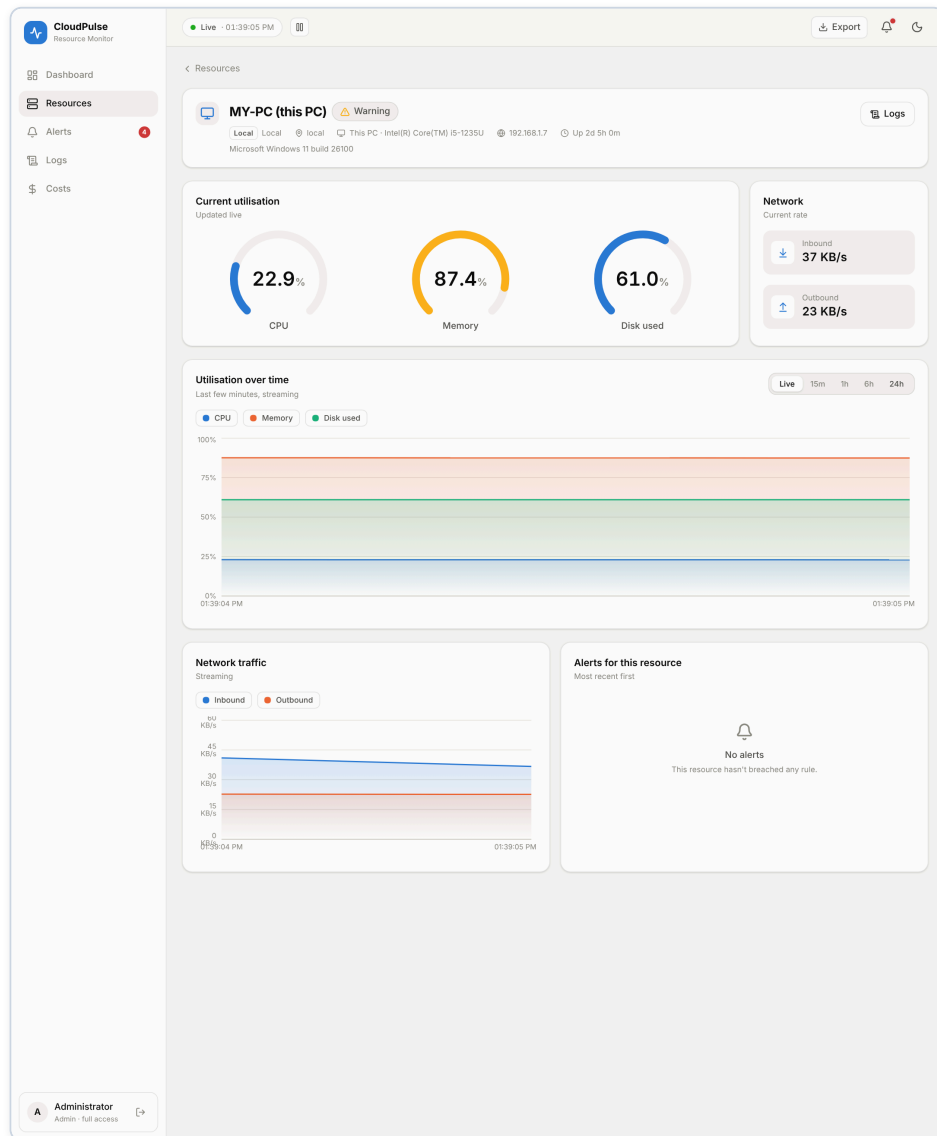


Figure 7 – Your computer, read live through OSHI

The row named "**(this PC)**" shows real values from your computer: CPU, memory, disk space used, network and Windows uptime. You can compare it with Task Manager. For safety it **cannot be stopped** from CloudPulse, so it has no Stop button.

Disk used means storage *space* used. Task Manager's "Disk" column shows disk *activity*, so the numbers are different. Both are correct.

Alerts

9.1 Pop-ups

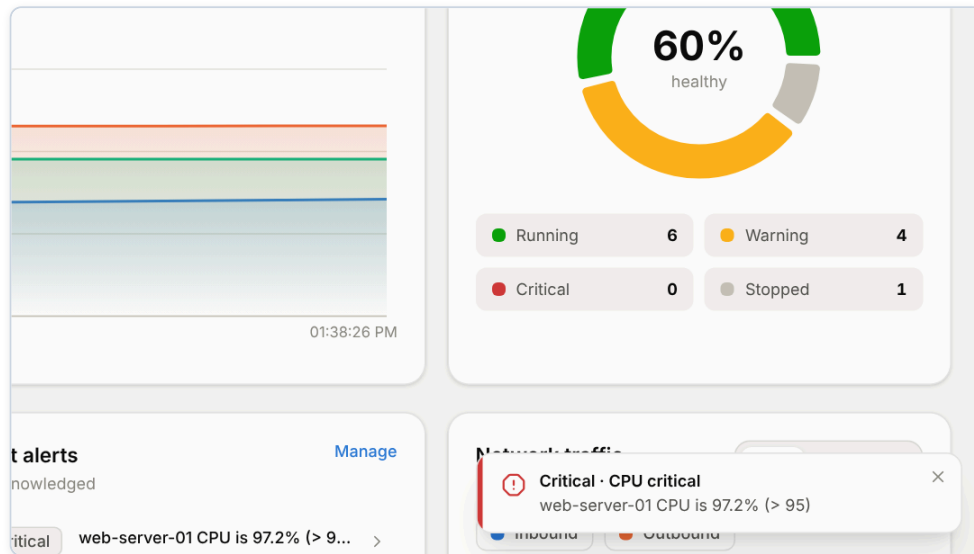


Figure 8 – A new critical alert appears at the bottom right

When a rule is broken, a pop-up appears on **every open screen** at once. Click × to close it. The alert stays on the Alerts page until it is acknowledged.

9.2 Active alerts

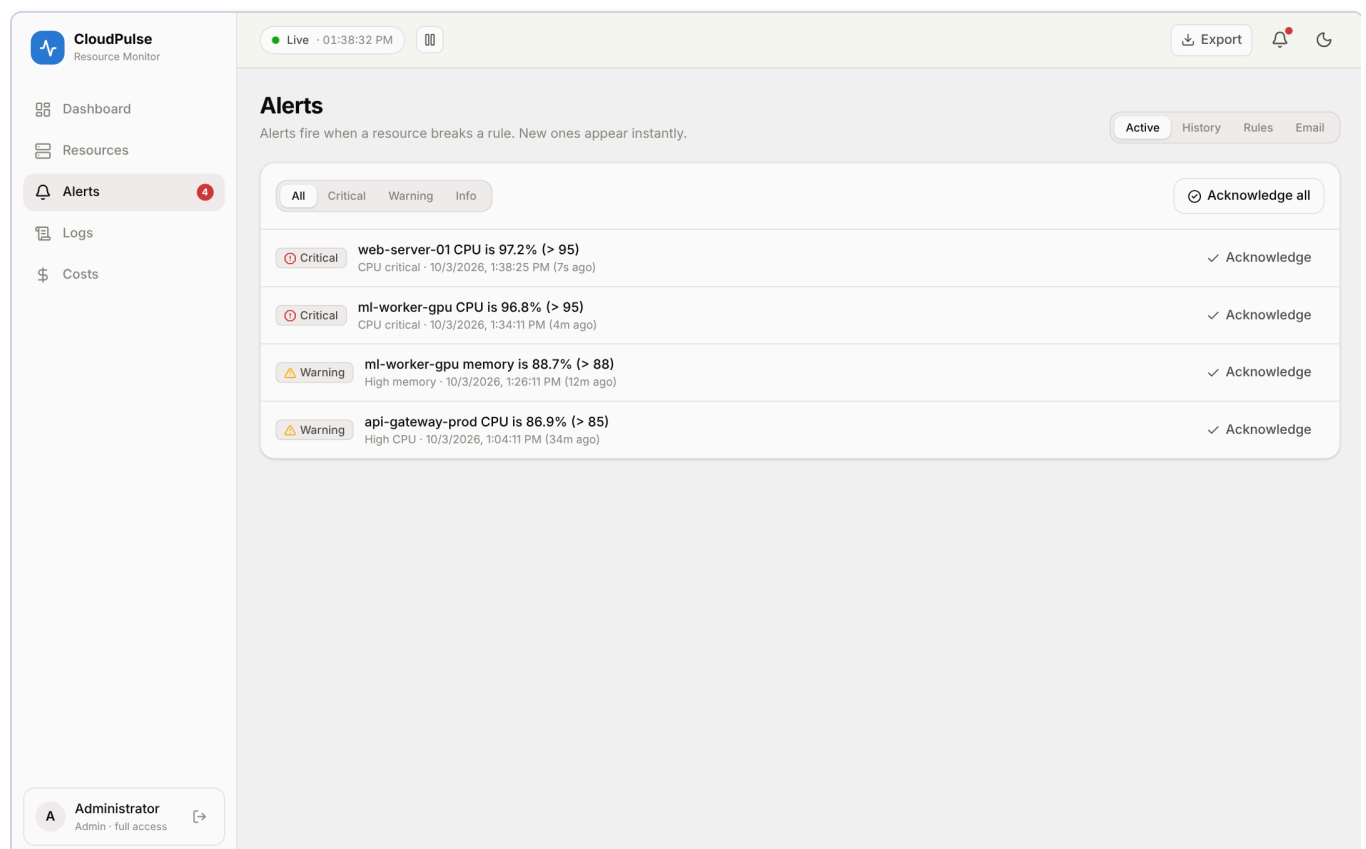


Figure 9 – Alerts page, Active tab

- Shows alerts that nobody has acknowledged yet.
- Filter by **All**, **Critical**, **Warning**, **Info**.
- Click **Acknowledge** on one alert, or **Acknowledge all**. Your username is saved with it.

9.3 History

The **History** tab shows all recent alerts, including acknowledged ones and who acknowledged them.

CloudPulse
Resource Monitor

Live · 01:38:32 PM

Export

Alerts

Alerts fire when a resource breaks a rule. New ones appear instantly.

Active **History** Rules Email

All Critical Warning Info

Critical	web-server-01 CPU is 97.2% (> 95) CPU critical · 10/3/2026, 1:38:25 PM (7s ago)	✓ Acknowledge
Critical	ml-worker-gpu CPU is 96.8% (> 95) CPU critical · 10/3/2026, 1:34:11 PM (4m ago)	✓ Acknowledge
Warning	ml-worker-gpu memory is 88.7% (> 88) High memory · 10/3/2026, 1:26:11 PM (12m ago)	✓ Acknowledge
Warning	api-gateway-prod CPU is 86.9% (> 85) High CPU · 10/3/2026, 1:04:11 PM (34m ago)	✓ Acknowledge
Critical	analytics-vm CPU is 96.3% (> 95) CPU critical · 10/3/2026, 12:38:11 PM (1h ago) · acknowledged by admin	
Warning	web-server-01 CPU is 87.4% (> 85) High CPU · 10/3/2026, 12:03:11 PM (1h ago) · acknowledged by viewer	
Warning	ml-worker-gpu memory is 89.1% (> 88) High memory · 10/3/2026, 11:08:11 AM (2h ago) · acknowledged by admin	
Warning	postgres-main CPU is 88.2% (> 85) High CPU · 10/3/2026, 10:38:11 AM (3h ago) · acknowledged by admin	
Warning	backup-server was stopped by admin System event · 10/3/2026, 10:18:11 AM (3h ago) · acknowledged by admin	

Administrator
Admin · full access

Figure 10 – Alert history

9.4 Alert rules (admin)

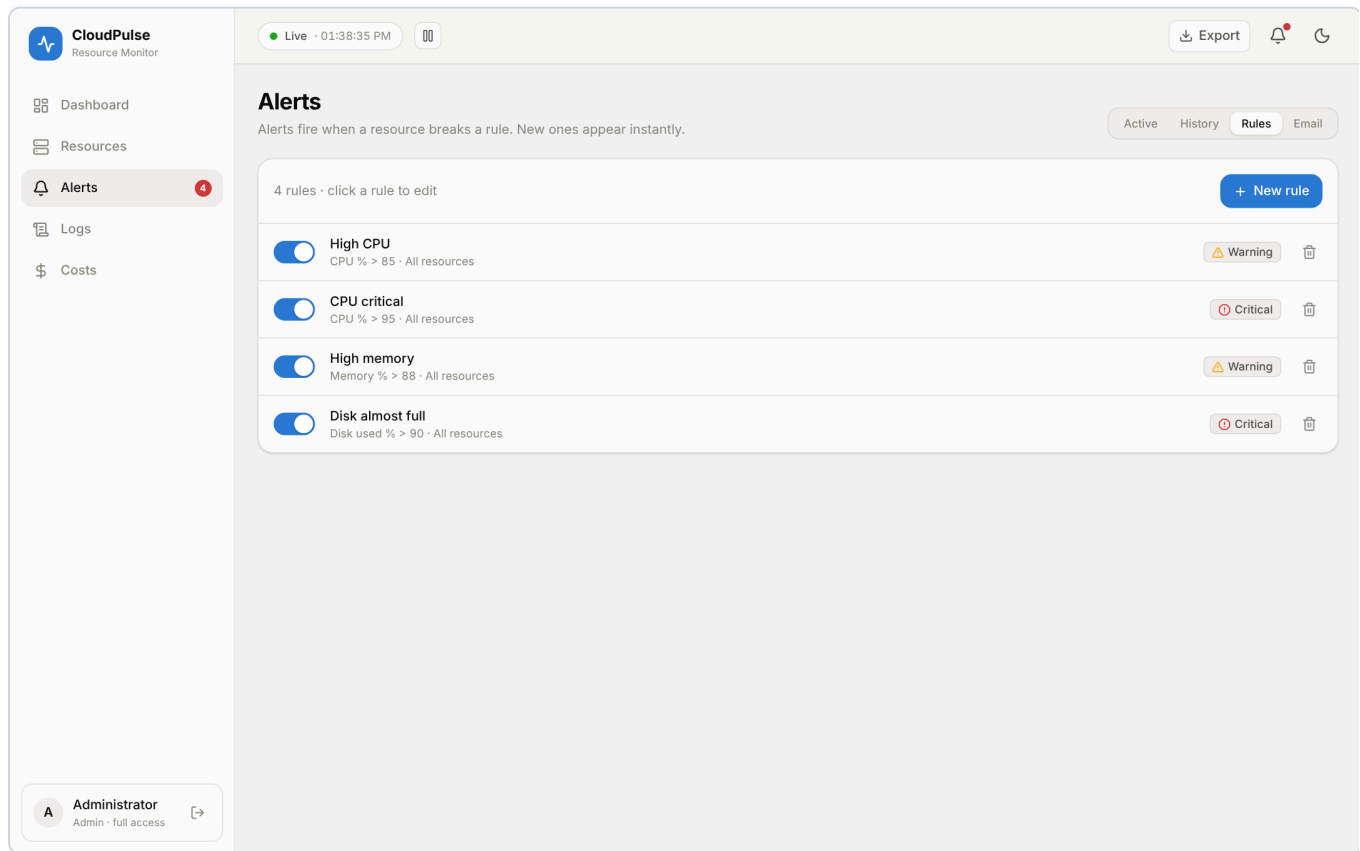


Figure 11 – Rules tab

CloudPulse starts with four rules:

Rule	Condition	Severity
High CPU	CPU > 85%	Warning
CPU critical	CPU > 95%	Critical
High memory	Memory > 88%	Warning
Disk almost full	Disk > 90%	Critical

Create a rule:

1. Click **New rule**.
2. Enter a **Name** (e.g. "Network spike").
3. Choose the **Metric** (CPU, memory, disk, network in, network out), the **Condition** (above / below) and the **Threshold** (a slider for %, or a number in KB/s for network).
4. Choose **Applies to**: all resources or one resource.
5. Choose the **Severity**: Info, Warning or Critical.
6. Click **Save rule**.

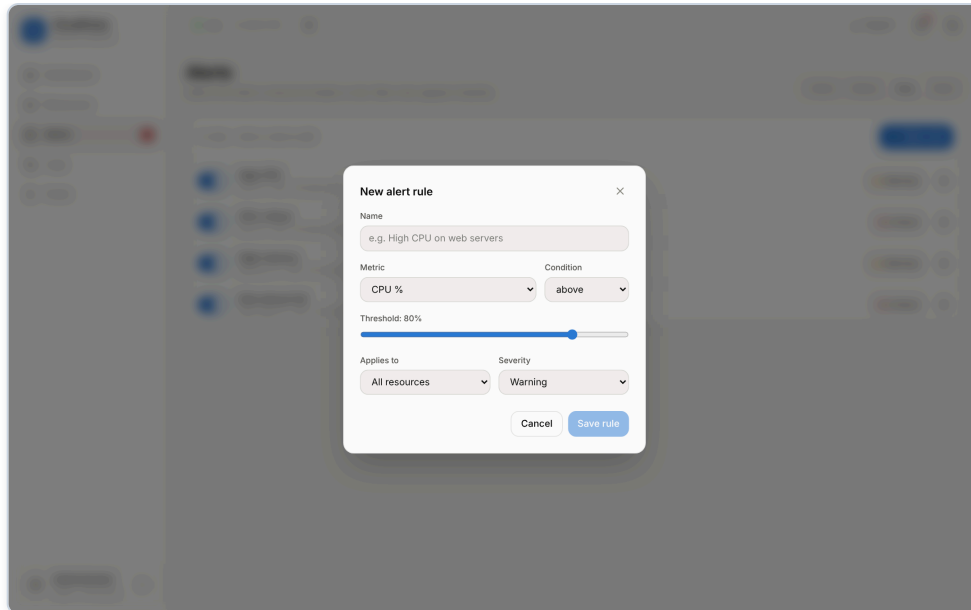


Figure 12 – New alert rule

- **Edit:** click anywhere on a rule.
- **Turn on/off:** use the switch on the left of the rule.
- **Delete:** the bin icon on the right.

Spam protection: the same rule on the same resource fires at most **once every 2 minutes**.

Email alerts

10.1 One-time setup on the server

1. In your Google Account turn on **2-Step Verification**.
2. Go to **Google Account → Security → App passwords**, create one (name it "CloudPulse") and copy the **16-letter password**.
3. In IntelliJ: **Run → Edit Configurations → CloudMonitorApplication → Environment variables**. Add:

```
MAIL_USERNAME=your.name@gmail.com
MAIL_PASSWORD=abcdefghijklmnop
```

Check there are **no spaces** before the names. Leave "Store as project file" unticked so the password never goes to GitHub.

4. Restart the backend. The Run window should show **Email alerts: ready**.

Never write the App password in the code or share it with anyone. If it leaks, delete it in Google Account and make a new one.

10.2 Email tab (admin)

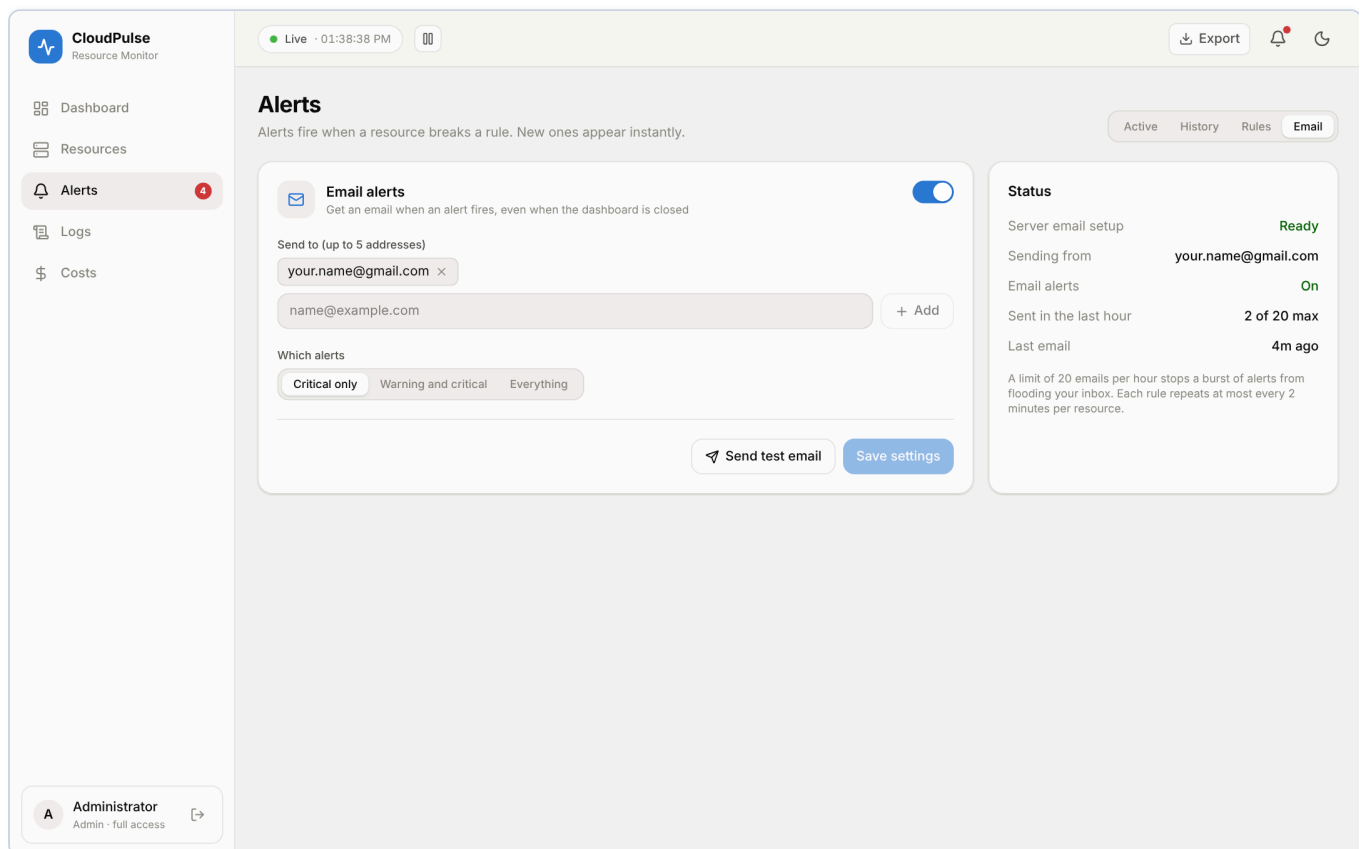


Figure 13 – Alerts → Email tab

1. Open **Alerts → Email**. "Server email setup: Ready" means step 10.1 worked.

2. Turn **Email alerts** on with the switch.
3. Add up to **5** addresses under **Send to** (type and click **Add**; × removes one).
4. Choose **Which alerts**: Critical only (recommended), Warning and critical, or Everything.
5. Click **Save settings**, then **Send test email** to check your inbox.

The **Status** box shows the sender, how many emails were sent in the last hour (limit **20 per hour**), when the last one was sent and any error.

Costs and budget

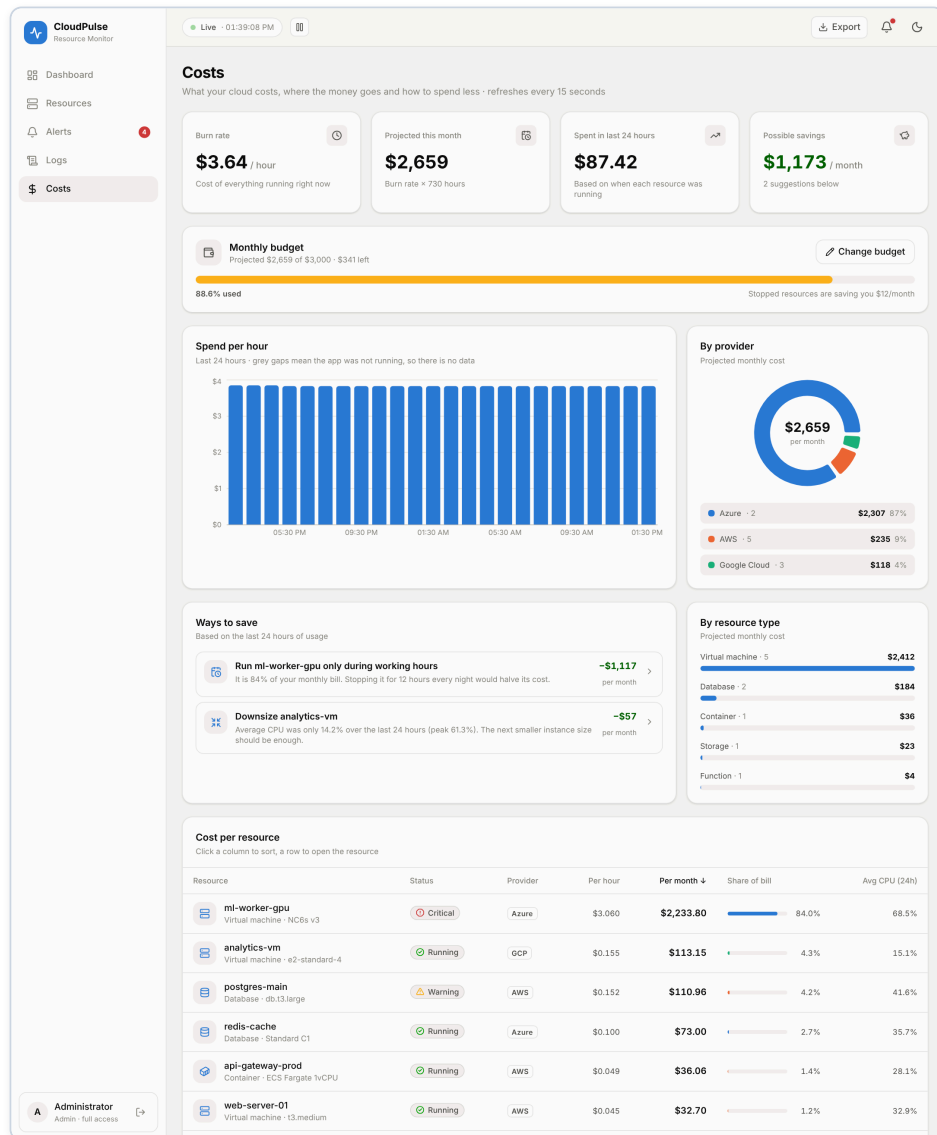


Figure 14 – Costs page

Area	Meaning
Burn rate	What all running resources cost per hour right now.
Projected this month	Burn rate × 730 hours (one month).
Spent in last 24 hours	What was actually spent in the last day.
Possible savings	Total of all savings tips per month.
Monthly budget / Budget used	How much of the budget the projected month uses.
Spend per hour	Cost of each of the last 24 hours. Grey gaps mean CloudPulse was not running then.
By provider / by type	Where the money goes.

Ways to save	Tips: <i>run only during working hours</i> (one resource is 40%+ of the bill) or <i>downsize</i> (CPU on average below 25%).
Cost per resource	Price per hour, per month, and share of the bill for each resource. Click a column to sort, a row to open the resource. Your PC is free and "Not billed".

Change the budget (admin)

1. Click **Change budget**.
2. Type the new monthly amount in USD.
3. Click **Save budget**.

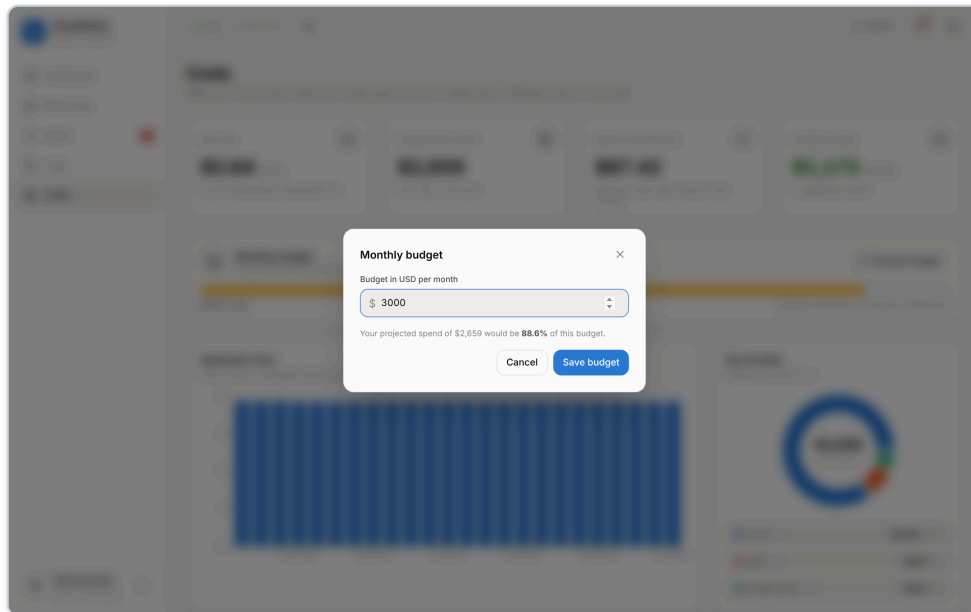


Figure 15 – Changing the monthly budget

Try it: stop the expensive **ml-worker-gpu** server and watch the burn rate fall from about \$3.64 to about \$0.58 per hour.

Logs

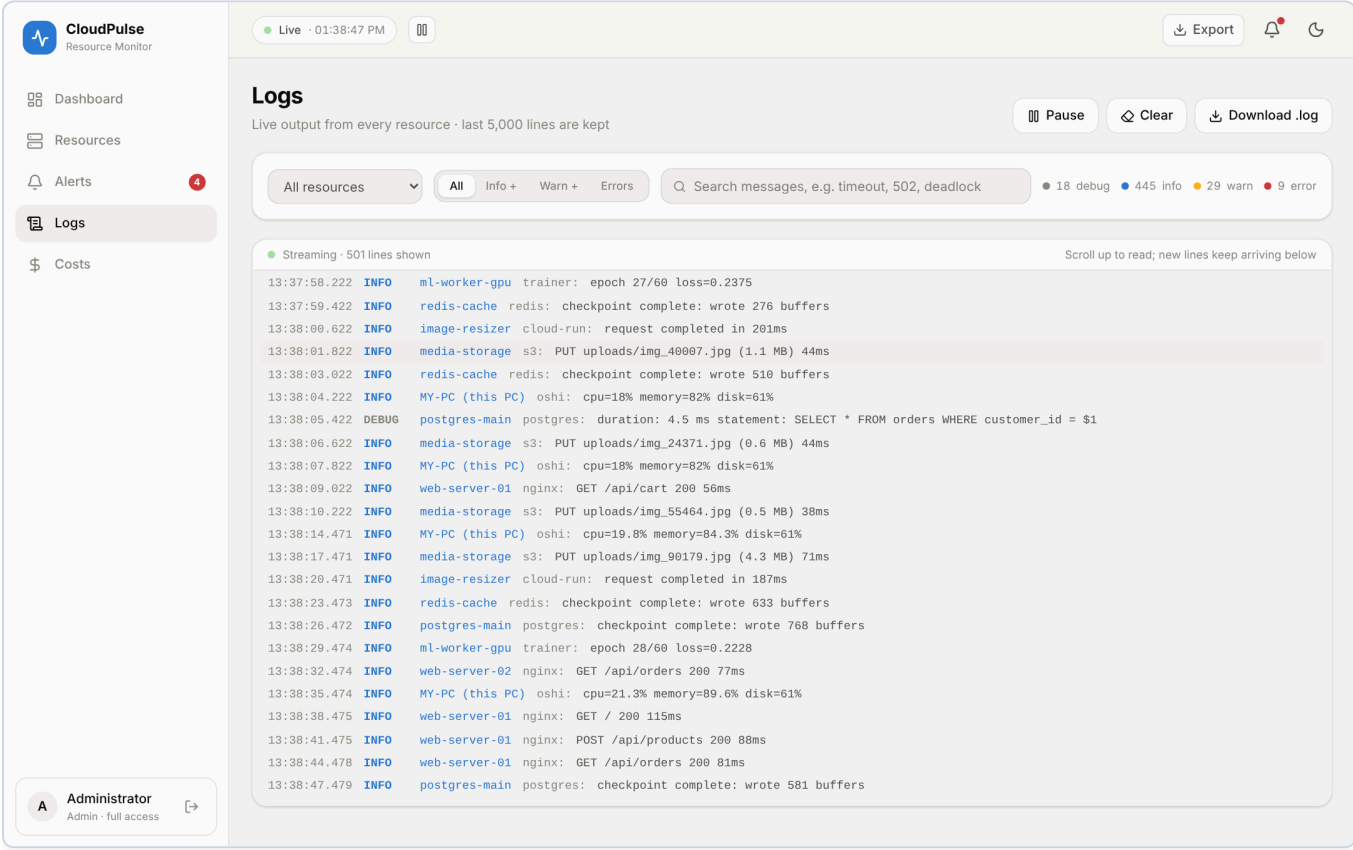


Figure 16 – Live logs

Each line shows: **time • level • resource • program • message**. New lines appear at the bottom automatically.

Control	Use
Resource list	Show one resource or all.
Level filter	Minimum level to show: DEBUG (everything), INFO, WARN, ERROR (only problems).
Search box	Find words, e.g. timeout , 502 , GPU .
Pause / Resume	Stop the stream to read. When resumed, it shows how many new lines were waiting.
Clear	Empties the screen only (logs stay on the server).
Download .log	Saves the lines that match your filters as a text file.

If you scroll to read, the view stops jumping to the bottom. The server keeps the newest **5,000** lines.

DEBUG details for developers **INFO** normal **WARN** something to watch **ERROR** a problem

Export reports

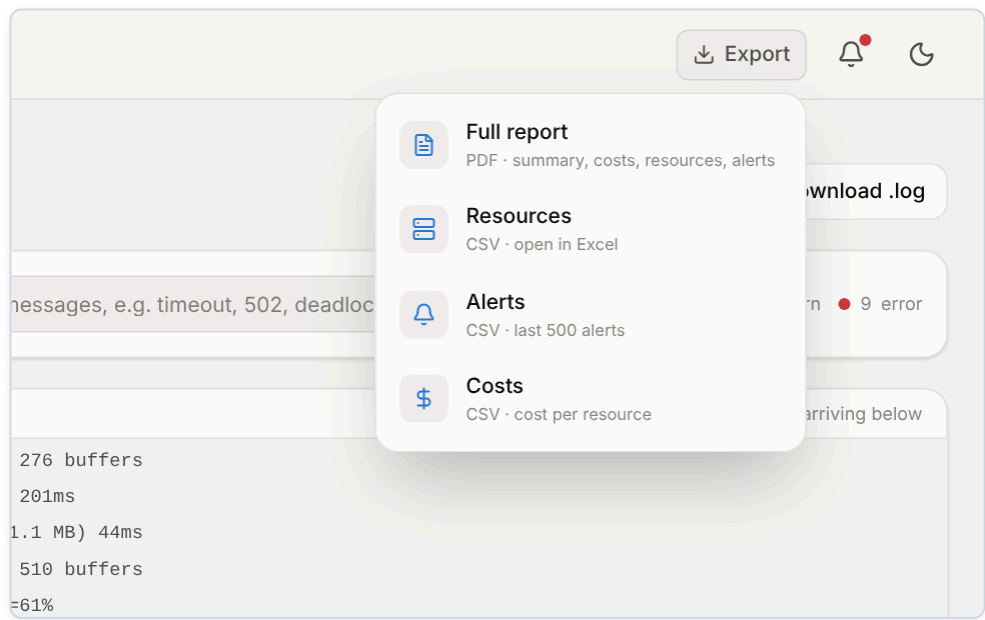


Figure 17 – Export menu (top bar)

Option	File	Contains
Full report	cloudpulse-report.pdf	Summary, costs, all resources and recent alerts
Resources	cloudpulse-resources.csv	All resources and their current values
Alerts	cloudpulse-alerts.csv	Last 500 alerts
Costs	cloudpulse-costs.csv	Cost of each resource

Click **Export** in the top bar and choose a file. It downloads into your browser's Downloads folder. Open CSV files in Excel.

Dark mode and phone

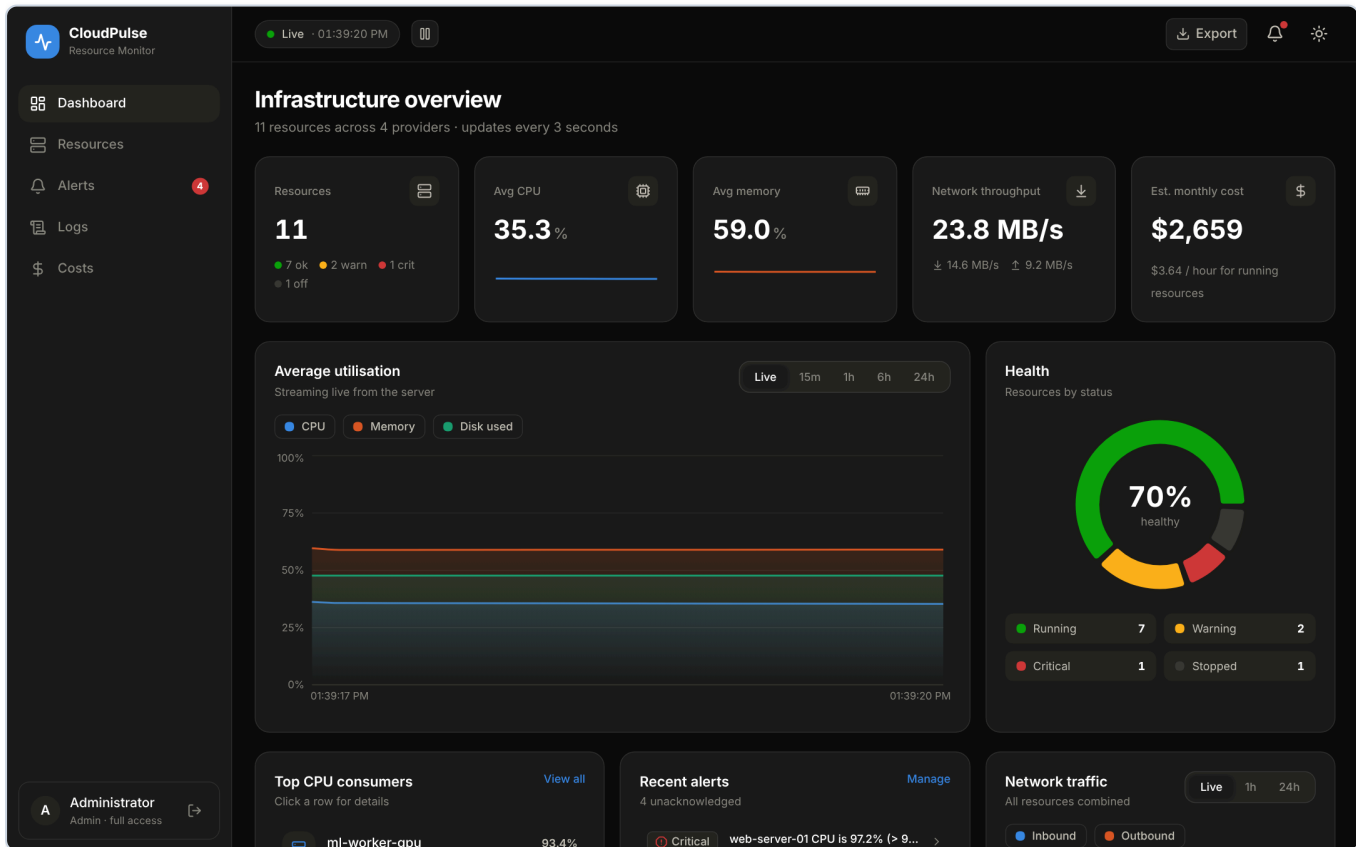


Figure 18 – Dark theme

Click the **moon/sun** icon in the top bar to switch theme. Your choice is remembered on that browser.

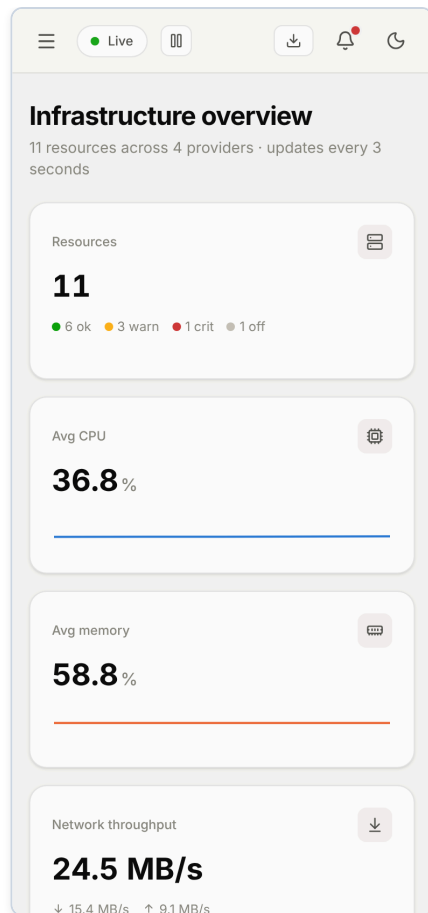


Figure 19 – On a phone

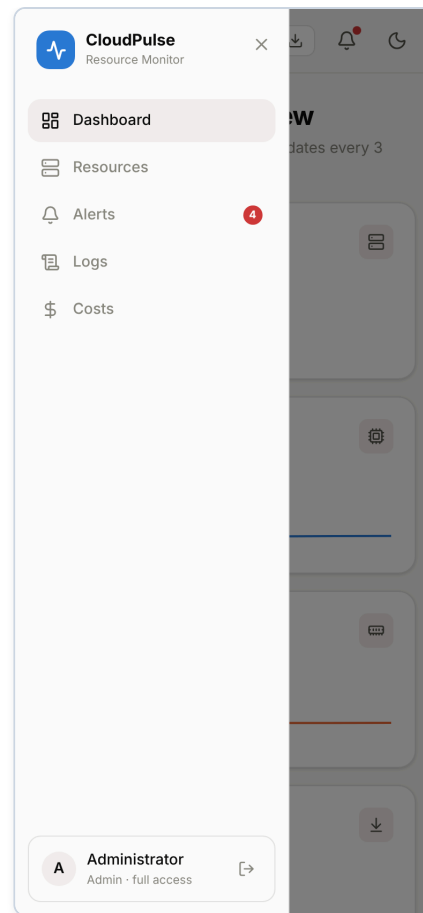


Figure 20 – Phone menu (☰)

On a small screen the menu hides behind the ☰ button at the top left. To open CloudPulse from a phone on the same Wi-Fi, the frontend must be started with network access (`npm run dev -- --host`) and you open `http://<your-PC-IP>:5173` .

Admin vs Viewer

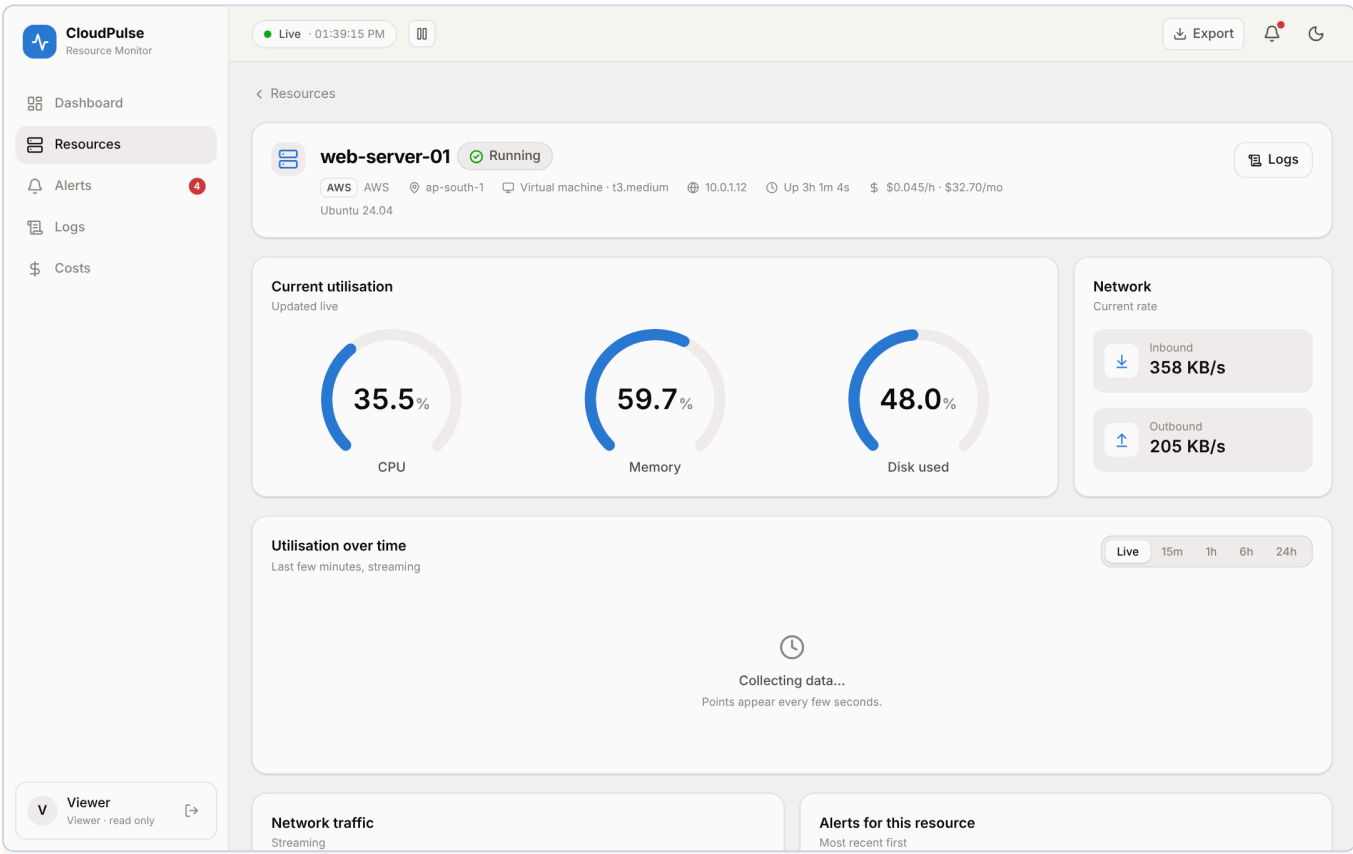


Figure 21 – Logged in as Viewer: no Start / Stop / Restart buttons

Feature	Admin	Viewer
Dashboard, Resources, details, charts	Yes	Yes
Logs, Costs	Yes	Yes
Acknowledge alerts	Yes	Yes
Export PDF / CSV	Yes	Yes
Start / stop / restart resources	Yes	No
Create / edit / delete rules	Yes	No
Change budget	Yes	No
Email settings, test email	Yes	No

These limits are checked by the server too, so a viewer cannot get around them.

Troubleshooting

Problem	Likely cause	What to do
Browser says "This site can't be reached" on localhost:5173	Frontend not running	Run <code>npm run dev</code> in the frontend folder.
Pages empty, "Offline", or Vite shows <code>ECONNREFUSED</code>	Backend not running	Start CloudMonitorApplication in IntelliJ, wait for "Started".
Backend error: <code>Access denied for user 'root'</code>	Wrong MySQL password	Fix <code>spring.datasource.password</code> or set <code>DB_PASSWORD</code> .
Backend error: <code>Communications link failure</code>	MySQL is not running	Start the MySQL80 service.
Backend error: <code>Port 8080 was already in use</code>	Backend already running (or another program)	Stop the other run in IntelliJ, or close the program using 8080.
Login says "Invalid username or password"	Typing mistake	Use admin / admin123 or viewer / viewer123 (lower case).
Sent back to login suddenly	8-hour login expired	Log in again.
Email tab shows "Not set up"	Environment variables missing or have spaces	Check Section 10.1, then restart the backend.
Test email fails with an authentication error	Wrong App password, or normal Gmail password used	Create a new App password and paste it again.
No emails for warnings	"Critical only" is selected	Choose "Warning and critical" in the Email tab.
Disk % differs from Task Manager	Different meaning (space vs activity)	Normal. See Section 8.3.
Charts are not moving	Live updates paused	Click the Pause/Resume button in the top bar.
IntelliJ shows old code after a change	File changed outside IntelliJ	File → Reload All from Disk.

Frequently asked questions

Are the cloud servers real?

No. The 10 cloud servers are simulated to behave like real ones (busy in the afternoon, quiet at night, sudden spikes). Your PC's data is real.

Will my data be lost when I close it?

No. Everything is in MySQL. Only the screen log lines older than 10 minutes are not kept after a restart.

How long is history kept?

24 hours. Older points are removed automatically.

How do I reset everything to the demo data?

Stop the backend, delete the database in MySQL (`DROP DATABASE cloudmonitor;`) and start the backend again.
This deletes all alerts, settings and history.

Can I add my own users?

Not from the screen in this version. The two demo users are created automatically.

Can two people use it at the same time?

Yes. Everyone sees the same live data and pop-ups at the same moment.

Why did I get only one email for a long problem?

Spam protection: the same rule repeats at most every 2 minutes per resource, and at most 20 emails are sent per hour.

Quick reference card

Start

1. MySQL running
2. IntelliJ → Run backend
3. `npm run dev`
4. Open localhost:5173

Logins

admin / admin123
viewer / viewer123
Valid for 8 hours

Addresses

Website: localhost:5173
Server: localhost:8080
MySQL: localhost:3306

Status colours

- Running < 75%
- Warning 75–89%
- Critical ≥ 90%
- Stopped

Limits

Updates every 3 s
History 24 h
Logs 5,000 lines
Email 20/hour, 5 recipients
Same alert every 2 min max

Where is...

Start/stop → Resources → open one
Rules → Alerts → Rules
Email → Alerts → Email
Budget → Costs
Reports → Export (top)